Customer Churn Analysis & Prediction Report for a Telecommunication Company (Python)

Project Type: Data Analysis & Machine Learning
Tools: Python, Pandas, NumPy, Seaborn, Scikit-learn
Industry: Telecommunications

Project Overview:

Telecom companies face significant revenue loss due to customer churn. In this project, I examined customer churn data from a telecommunications company to uncover the main factors driving customer churn and develop machine learning models to predict customers at risk of leaving.

The goal was to transform raw customer data into actionable insights that can support **targeted retention strategies** and improve long-term customer value.

Business Problem:

- Customer churn directly impacts recurring revenue.
- Most churn is identifiable **before it happens**.
- The business needed:
  - Early churn detection
  - Clear understanding of why customers leave.
  - Data-backed retention decisions.

Project Objectives

- Analyse customer behaviour and churn patterns.
- Identify the strongest churn drivers.
- Build predictive models to flag at-risk customers.
- Deliver insights in a **business-friendly format**.

Dataset Overview

- **7,043 customers**
- **21 features**, including:
  - Demographics (gender, senior citizen, dependents)
  - Services subscribed (internet, streaming, support)
  - Billing & payments (charges, contracts, payment method)
  - Target variable: **Churn (Yes / No)**

Objectives:

- Predict customer churn using machine learning models (Logistic Regression, Random Forest).
- Identify key features influencing churn (e.g., contract type, tenure, charges, internet service).
- Compare model performance using metrics like accuracy, precision, recall, F1-score, and AUC.
- Provide business insights to guide customer retention initiatives.

Tools & Technologies

- Python (Pandas, NumPy)
- Data visualization: Matplotlib, Seaborn
- Machine Learning: Scikit-learn
- Class imbalance handling: SMOTE
- Models: Logistic Regression & Random Forest
- Model deployment (.pkl files)

Workflow:

- Data cleaning and pre-processing
- Performed exploratory data analysis (EDA)
- Feature engineering and class imbalanced handled
- Built and evaluated multiple machine learning models.
- Model training and evaluation
- Compared models using Accuracy, Precision, Recall, and ROC-AUC.
- Saved trained models for deployment.
- 5. Business insights

Key Performance Indicators (KPIs):

- Churn Rate (%) → Percentage of customers leaving.
- Recall (Churn class) → Ability to correctly identify churners.
- Precision (Churn class) → Accuracy of churn predictions.
- AUC (ROC Curve) → Overall discriminative power of models.
- Customer Lifetime Value (CLV) → Estimated revenue impact of churn.

Modelling Approach

Model	Purpose
Logistic Regression	Clear interpretability for business
Balanced Logistic Regression	Improved churn recall
Random Forest	Captures complex behaviour patterns

Modelling Results:

- **Best Accuracy:** ~79%
- **Best ROC-AUC:** ~0.83
- Balanced Logistic Regression improved churn recall to **~79%**
- Logistic Regression showed the **best generalization** in cross-validation.

Logistic Regression delivered strong interpretability and slightly higher AUC in cross-validation, while Random Forest captured nonlinear patterns and interactions for robust performance.

Key Business Insights:

- Customers on **month-to-month contracts** churn significantly more than long-term contract customers. **High monthly charges** strongly increase churn likelihood.
- Customers with **longer tenure** are far less likely to churn, while **Short-tenure customers** are most at risk.
- Model Performance: Logistic Regression slightly outperformed Random Forest in AUC (0.83 vs 0.82) and cross-validation (0.85 vs 0.83).
- Churn Drivers: Fiber optic internet, high charges, paperless billing, and electronic check payments are strong churn predictors. However, **Fiber optic internet users** churn more than DSL users.
- Class Imbalance: Both models predict non-churners well but struggle with churn recall (~50%).
- Retention Focus: Improving recall is more valuable than overall accuracy; better to flag potential churners even at the cost of false positives.

Business Value:

This solution enables businesses to:

- Identify customers most likely to churn.
- Design targeted retention campaigns.
- Reduce revenue loss.
- Make data-driven customer engagement decisions.

Note: This is a portfolio project demonstrating real-world analytics skills. The workflow is fully adaptable for client-side deployment.

Final Deliverables

- Clean, well-documented Jupyter Notebook
- Churn prediction models (.pkl)
- Business insights & recommendations
- README

Conclusion:

- Logistic Regression provides interpretability and slightly stronger generalization.
- Random Forest offers feature importance and robust non-linear modelling.
- Combining both models (ensemble or stacked approach) can balance interpretability and predictive strength.
- Business teams should focus on targeted retention campaigns for high-risk groups identified by the models.

Outcome

This project demonstrates my ability to:

- Translate complex data into business insight.
- Build production-ready ML solutions.
- Communicate findings to non-technical stakeholders.